

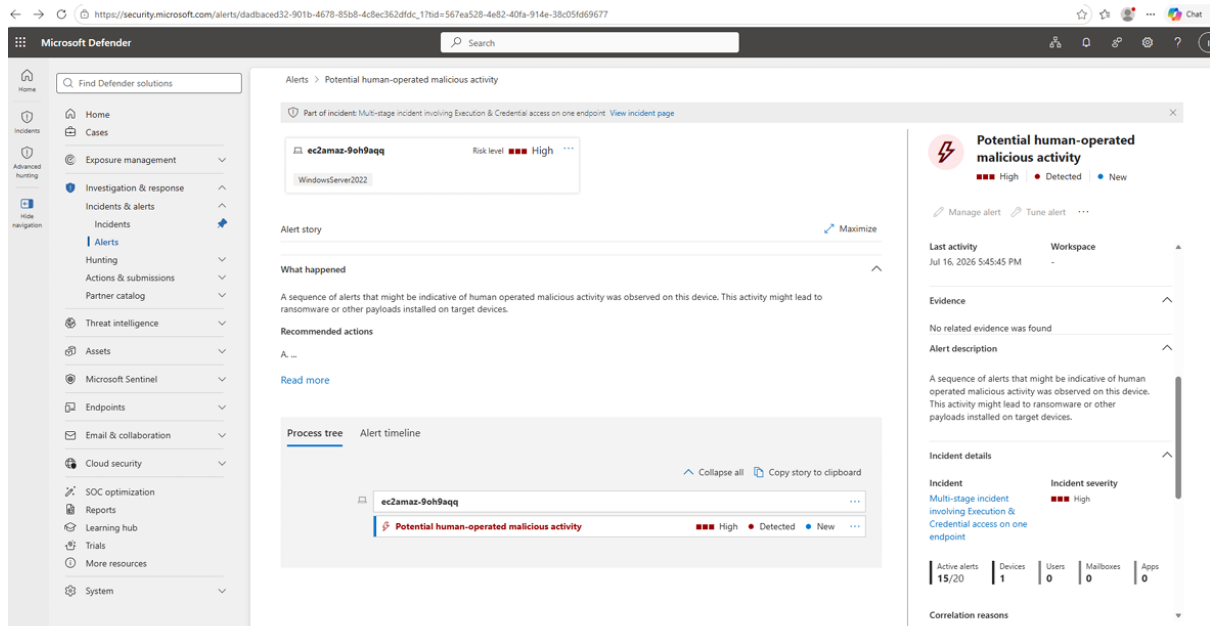
Alert: Potential human-operated malicious activity

Endpoint: ec2amaz-9oh9aqq

User: No user

OS: WindowsServer2022

Verdict: True positive



Why was the alert triggered?

Microsoft Defender for Endpoint generated a high severity behavioural alert indicating potential human-operated malicious activity. The alert was produced by Machine Learning after correlating multiple suspicious behaviours on device **ec2amaz-9oh9aqq**

The detection name "Potential human-operated malicious activity" is one of Microsoft's behavioural detections that usually appears during hands-on-keyboard attacks rather than commodity malware.

Investigation Summary

Microsoft Defender for Endpoint generated a **high** severity alert for **Potential human-operated malicious activity** on **ec2amaz-9oh9aqq (Windows Server 2022)**. The detection was generated by Microsoft's behavioural machine learning engine after correlating multiple suspicious activities indicative of a hands-on-keyboard attack. The incident was part of **multi-stage incident involving Execution and Credential Access**, with **3 correlated alerts** occurring over approximately 40 minutes.

Review of the alert indicates the activity was **detected**, however there is no evidence that Microsoft Defender blocked or remediated the malicious actions. The alert does not include a process tree or specific evidence identifying the executed process, that is why other related alerts were investigated to determine the initial access vector, attack sequence, and scope of compromise. Given the behavioural nature of the detection and multiple correlated alerts, the activity is assessed as a **True Positive** with a high likelihood of interactive attacker activity.

MITRE ATT&CK Tactics: Execution, Credential Access, Privilege Escalation, Persistence, Defence Evasion, Lateral Movement (to be confirmed during investigation).

Remediation Steps

1. **Isolated the affected endpoint** (ec2amaz-9oh9aqq) using Microsoft Defender for Endpoint to prevent further attacker activity while preserving forensic evidence.
2. **Reviewed the incident timeline** and all 3 correlated alerts to identify the initial execution point, attack chain, affected accounts, and impacted assets and **isolated** the affected hosts.
3. **Investigated endpoint activity**, including process execution, PowerShell and Command Prompt activity, registry changes, scheduled tasks, services, file creation/deletion, and network connections.
4. **Collected forensic evidence** such as memory capture (if appropriate), event logs, EDR telemetry, and relevant system artefacts to determine the extent of compromise.
5. **Terminated malicious processes** and removed any identified malicious files or persistence mechanisms, including scheduled tasks, services, registry Run keys, start-up items, and WMI event subscriptions.
6. **Updated and hardened the endpoint** by requesting **services team** for getting latest Windows security updates, ensuring Microsoft Defender signatures and platform are current, enabling Tamper Protection, and reviewing firewall and application control policies.
7. **Continued monitoring** for recurring execution, credential access, privilege escalation, persistence, and lateral movement activities, and tune detection rules based on investigation findings.